NPS Guide Copilot

An End-to-End LLMOps Pipeline for a Grounded Multi-Agent Assistant over Fast-Changing National Park Data

Weekly Status Update · Submitted for class · Oct 28, 2026

1. Project Summary

NPS Guide Copilot sits on top of **NPS Guide**, a deterministic U.S. (and nearby) national-park trip planner (Next.js). The Copilot understands natural-language trip requests, orchestrates tools, and explains plans—while the website engine still does the planning. Focus areas: multi-agent orchestration, PromptOps, evaluation gates in CI, and monitoring over fast-changing NPS alert data.

Goal this semester: ship a grounded LLMOps system—contracts, local + cloud models, replay evals in CI, nightly alert extraction with human-in-the-loop closure proposals, and a chat panel on the trip planner—demo-ready by Dec 10, 2026.

Recent updates (Oct 20–28):

- **Live NPS alerts loop:** nightly fetch → model extract → weekly quality canary → propose closures for human review → publish live closure JSON for the site (fixed canary enable/disable bug).
- **Copilot tools & UX:** `is_open` / `drive_time` / `search`; streaming progress UI; multi-park trips; LangGraph-shaped pipeline hooks.
- **Coverage:** all **64** U.S. national parks on the site; AllTrails popularity for hikes; Tesla Supercharger / destination maps per park.
- **Ops:** Copilot **v0.3.0**; Hugging Face Spaces prep (auth + rate limits); one-click local launcher; admin architecture diagram + English admin for instructors.
- **Data quality (Huiling):** ~85 seasonal campgrounds added to `lodging-seasons.ts` so winter “camping” no longer picks summer-only camps (PR #66).

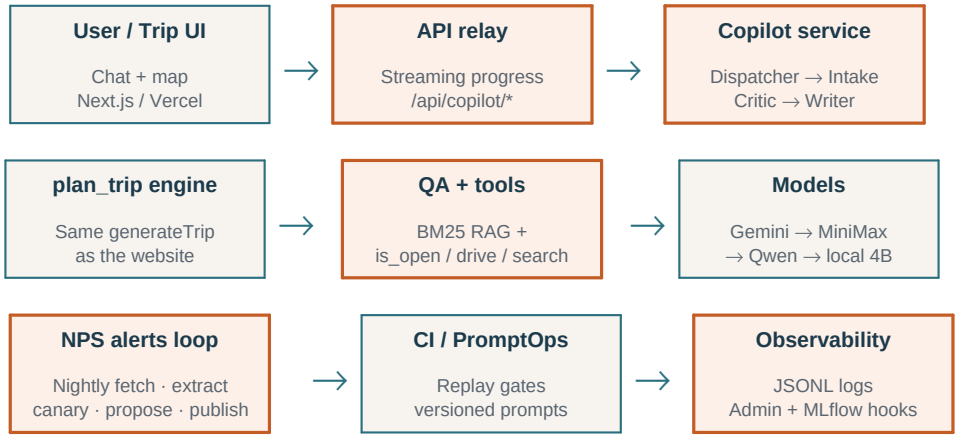
2. Team Members

Name	Role	Notes / changes this week
Jun Zhai (GitHub: momomojun)	Owner · data, agent, CI/CD, architecture	Led alerts loop, tools, parks coverage, admin diagram, releases
Tianyue Chen (GitHub: JadenChen10)	Eval, metrics, monitoring (planned)	Collaborator on repo; proposed park-safety module (O63, not started)
Huiling Fu (GitHub: huilingfu95-source)	Site / lodging data; deployment assist	Delivered campground open seasons (PR #66); first teammate-owned data PR

Change: Huiling moved from “placeholder teammate” to an active data owner (campground seasons). Proposal §2 should still be updated to name her and lock eval vs. deploy ownership for Tianyue.

3. Architecture Diagram

Updated system view (highlights this week in **orange**): user chat → Next.js relay (streaming) → Copilot (FastAPI / LangGraph) → deterministic `plan_trip` engine; parallel QA tools + nightly NPS alert pipeline.



Also on the site this week: lodging/campground seasons feed the same planner; Tesla charging pages are separate park data, not on the Copilot critical path.

4. Tech Stack

Layer	Technologies	Updates this week
Frontend	Next.js, React, TypeScript, MapLibre, Vercel	Progress UI; Tesla maps; EN admin; local launcher
Copilot	Python, FastAPI, LangGraph, Ollama	Tools; streaming; HF Spaces prep; v0.3.0
Models	Qwen3.5 4B (default), Gemini Flash Lite, MiniMax, Qwen cloud	Free-first failover chain; alerts canary weekly
Planning data	TS data packs, lodging-seasons, closures, AllTrails	+85 camp seasons; 64 parks; AllTrails ranks
LLMOps	GitHub Actions, GHCR, replay evals, PromptOps, JSONL / admin	Alerts propose/publish; canary fix; architecture diagram

5. Technical Demo (≤ 5 min)

This update includes a short demo slot. Suggested script (live laptop: site + Copilot via one-click launcher):

- **0:00–0:45 — Context:** show admin architecture diagram (`/en/admin`)—models vs. deterministic `plan_trip`.
- **0:45–2:15 — Chat plan:** "3 days Yosemite from Fresno in July, camping." Show streaming progress UI → itinerary lands on the planner.
- **2:15–3:30 — Grounding:** winter contrast—"January camping at Crater Lake / Rainier"—camps closed; falls back to open lodging (camp seasons).
- **3:30–4:30 — Fast-changing data:** open a recent `nps-alerts` issue / live closures path; mention weekly canary.
- **4:30–5:00 — Optional:** park Tesla charging map, or ask Copilot "is Tioga Road open mid-May?" via `is_open`.

Backup if GPU/cloud is down: replay a recorded eval turn + screenshots of the architecture diagram and PR #66 tests.

6. Plan for Next Week

- **Proposal submission** (course: week of Oct 29)—finalize PDF, name all three teammates and roles.
- **Alerts v2:** label unseen nightly alerts; tighten extraction; keep human review for `closures.ts`.
- **Hosted Copilot:** finish Hugging Face Space deploy so the public Vercel site can show chat without a local GPU.
- **Eval growth:** start holdout-v2 (multi-turn, conflicts, mixed language)—write expectations before first run.
- **Site polish:** bookings page layout when few entries (leftover from campground brief); optional safety module kickoff (Tianyue).

7. Blockers / Risks

Item	Impact	Support needed
Free-tier model quotas (MiniMax ~50/day)	Canary + nightly extract compete for calls	Keep canary weekly; document fallback chain
Copilot not on public Vercel yet	Chat hidden in production demo	Approve HF Space + secrets; or temporary demo laptop
Teammate ownership still thin in git history	Course "team collaboration" criterion	Lock P2 roles; Tianyue owns an eval/monitoring PR
Alert extraction false positives	Wrong closures remove popular stops	Strict+maybe tiers; human PR for closures.ts
tsc AppRoutes noise on Tesla page	Local typecheck noise unrelated to seasons PR	Regenerate Next types / small follow-up fix

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Repository: https://github.com/momomojun/Team · **Live site:** https://team-peach-five.vercel.app · **Campground seasons PR:** https://github.com/momomojun/Team/pull/66

Sources: PROJECT.md, GOALS.md, COPILOT.md, timeline entries 2026-10-20 through 2026-10-27.